A Priori Error Analysis

Galerkin orthogonality, best approximation, and interpolation

Lecture outline

The finite element computation gives u_h. We now ask how u_h compares with the exact weak solution u.

1. State the continuous and discrete variational problems.
2. Introduce the L^2, H^1, and energy measures of error.
3. Derive Galerkin orthogonality and show how it leads to best approximation.
4. Prove Céa's lemma.

The next lecture introduces finite element interpolation and uses it to obtain error estimates in terms of the mesh size h.

Continuous and discrete problems

Let

$$\mathcal{V}_0 = H^1_{0,\Gamma_D}(\Omega), \qquad a(w,v) = \int_\Omega k\,\nabla w \cdot \nabla v \,\mathrm{d}x.$$

The continuous problem is

$$\boxed{\text{find } u \in \mathcal{V}_0 : \quad a(u,v) = \ell(v) \quad \forall v \in \mathcal{V}_0.}$$

For a conforming finite element space $\mathcal{V}_{h,0} \subset \mathcal{V}_0$, the discrete problem is

$$\boxed{\text{find } u_h \in \mathcal{V}_{h,0} : \quad a(u_h,v_h) = \ell(v_h) \quad \forall v_h \in \mathcal{V}_{h,0}.}$$

The two problems use the same bilinear and linear forms. Only the spaces differ.

Assumptions for the analysis

We assume throughout this lecture that

- Ω is a bounded Lipschitz domain and Γ_D has positive boundary measure;
- $0 < k_0 \leq k(\boldsymbol{x}) \leq k_1$ almost everywhere in Ω;
- $\mathcal{T}_h$ is conforming and resolves the boundary partition;
- $\mathcal{V}_{h,0}$ is a continuous Lagrange space of degree $p \geq 1$ with homogeneous values on Γ_D; and
- the forms a and ℓ are evaluated exactly.

If quadrature or boundary approximation changes the discrete forms, the orthogonality relation contains an additional consistency error. We do not consider that case in this lecture.

Continuity and coercivity

The analysis uses two estimates:

$$|a(w,v)| \le M \|w\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}, \qquad w, v \in \mathcal{V}_0,$$

and

$$a(v,v) \ge \alpha \|v\|^2_{H^1(\Omega)}, \qquad v \in \mathcal{V}_0.$$

For scalar diffusion,

$$M = k_1, \qquad \alpha = \frac{k_0}{1 + C_P^2}, \qquad \frac{M}{\alpha} = \frac{k_1}{k_0}(1 + C_P^2).$$

The constants depend on the coefficient and the domain, but not on the mesh size h or polynomial degree p.

Existence and stability of the discrete solution

Because $\mathcal{V}_{h,0}$ is a finite-dimensional subspace of $\mathcal{V}_0$, the same continuity and coercivity estimates hold on $\mathcal{V}_{h,0}$.
The Lax–Milgram theorem therefore gives a unique finite element solution and

$$\|u_h\|_{H^1(\Omega)} \leq \frac{1}{\alpha} \|\ell\|_{\mathcal{V}_0^*} .$$

Coercivity also gives a positive-definite stiffness matrix. For any nonzero coefficient vector $\boldsymbol{V}$ representing $v_h \in \mathcal{V}_{h,0}$,

$$\boldsymbol{V}^T \boldsymbol{K}_{ff} \boldsymbol{V} = a(v_h, v_h) \geq \alpha \|v_h\|^2_{H^1(\Omega)} > 0.$$

When a is symmetric, the reduced stiffness matrix $\boldsymbol{K}_{ff}$ is symmetric positive definite.

Measures of the error

Define

$$e = u - u_h.$$

Because $u, u_h \in \mathcal{V}_0$, the error also belongs to $\mathcal{V}_0$.

$$\|e\|_{L^2(\Omega)} \quad \text{measures error in the solution values,}$$

$$|e|_{H^1(\Omega)} = \|\nabla e\|_{L^2(\Omega;\mathbb{R}^d)} \quad \text{measures error in the gradient,}$$

$$\|e\|_{H^1(\Omega)} = \left(\|e\|^2_{L^2(\Omega)} + |e|^2_{H^1(\Omega)}\right)^{1/2} \quad \text{measures both.}$$

For diffusion, the gradient error determines the flux error. For elasticity, it determines the strain error.

Energy norm

Suppose a is symmetric, continuous, and coercive on $\mathcal{V}_0$. Define

$$\boxed{\|v\|_a = a(v,v)^{1/2}.}$$

This is a norm because a is an inner product on $\mathcal{V}_0$.
For scalar diffusion,

$$\|v\|_a^2 = \int_\Omega k\,|\nabla v|^2 \,\mathrm{d}x.$$

The coefficient bounds give

$$\sqrt{k_0}\;|v|_{H^1(\Omega)} \le \|v\|_a \le \sqrt{k_1}\;|v|_{H^1(\Omega)}\,.$$

The energy norm weights the gradient by the material coefficient k.

Energy norm and the H^1 norm

On $\mathcal{V}_0$, the Poincaré inequality gives

$$\|v\|^2_{H^1(\Omega)} \leq (1 + C_P^2)\, |v|^2_{H^1(\Omega)}\,.$$

Combining this with the coefficient bounds gives

$$\boxed{\sqrt{\frac{k_0}{1 + C_P^2}}\ \|v\|_{H^1(\Omega)} \leq \|v\|_a \leq \sqrt{k_1}\ \|v\|_{H^1(\Omega)}\,.}$$

Thus convergence in the energy norm is equivalent to convergence in the H^1 norm on $\mathcal{V}_0$. The numerical values of the two norms need not be equal.

Energy norm for an axial bar

For an axial bar with $v(0) = 0$,

$$a(w, v) = \int_0^L AE\, w' v' \, \mathrm{d}x, \qquad \|e\|_a^2 = \int_0^L AE\, (e')^2 \, \mathrm{d}x.$$

The energy norm weights the strain error by the axial rigidity. When $AE = 1$,

$$\|e\|_a = |e|_{H^1(0,L)}.$$

Because $e(0) = 0$, the Poincaré inequality makes this seminorm equivalent to the full $H^1(0, L)$ norm.

Without an essential boundary condition, a nonzero constant displacement has zero strain and zero value of $a(v, v)$. The energy expression is then only a seminorm until rigid translations are removed.

Galerkin orthogonality

Every $v_h \in \mathcal{V}_{h,0}$ is also an admissible test function in the continuous problem. Hence

$$a(u, v_h) = \ell(v_h), \qquad a(u_h, v_h) = \ell(v_h).$$

Subtracting the discrete equation from the continuous equation gives

$$\boxed{a(u - u_h, v_h) = 0 \qquad \forall v_h \in \mathcal{V}_{h,0}.}$$

This conclusion uses conformity, $\mathcal{V}_{h,0} \subset \mathcal{V}_0$, and the same forms a and ℓ in the continuous and discrete problems. Symmetry of a is not needed for this result.

Orthogonality for a symmetric problem

If a is also symmetric, then

$$(w, v)_a := a(w, v)$$

is an inner product on $\mathcal{V}_0$, with induced norm

$$\|v\|_a = a(v, v)^{1/2}.$$

With $e = u - u_h$, Galerkin orthogonality becomes

$$(e, v_h)_a = 0 \qquad \forall v_h \in \mathcal{V}_{h,0}.$$

Thus the error is orthogonal to every function in the finite element space, where orthogonality is measured by the inner product a.

Pythagorean identity in the energy norm

For any $v_h \in \mathcal{V}_{h,0}$,

$$u - v_h = (u - u_h) + (u_h - v_h).$$

The second term belongs to $\mathcal{V}_{h,0}$, so Galerkin orthogonality gives

$$a(u - u_h, u_h - v_h) = 0.$$

Using bilinearity and symmetry of a,

$$\begin{aligned} \|u - v_h\|_a^2 &= a\big((u - u_h) + (u_h - v_h), (u - u_h) + (u_h - v_h)\big) \\ &= \|u - u_h\|_a^2 + 2a(u - u_h, u_h - v_h) + \|u_h - v_h\|_a^2 . \end{aligned}$$

The middle term is zero by Galerkin orthogonality. Therefore,

$$\boxed{\|u - v_h\|_a^2 = \|u - u_h\|_a^2 + \|u_h - v_h\|_a^2 .}$$

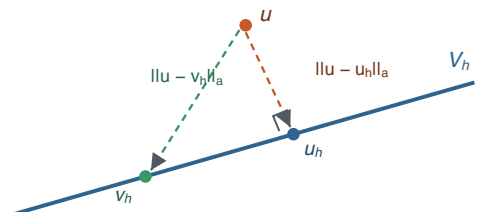
Best approximation in a selected norm

Let V_h be a finite-dimensional subspace of a normed vector space V. The best-approximation error is

$$E_V(u; V_h) = \min_{v_h \in V_h} \|u - v_h\|_V .$$

It is the shortest distance, measured with $\|\cdot\|_V$, from u to a function in V_h.

Changing the norm changes how distance is measured and can change which function in V_h is closest to u.



The Galerkin solution is the energy-norm best approximation

For any $v_h \in \mathcal{V}_{h,0}$, the Pythagorean identity gives

$$\|u - v_h\|_a^2 = \|u - u_h\|_a^2 + \|u_h - v_h\|_a^2 .$$

Since the last term is nonnegative,

$$\|u - v_h\|_a \geq \|u - u_h\|_a \qquad \forall v_h \in \mathcal{V}_{h,0}.$$

Equality holds when $v_h = u_h$. Therefore,

$$\boxed{\|u - u_h\|_a = \min_{v_h \in \mathcal{V}_{h,0}} \|u - v_h\|_a .}$$

The finite element solution is the function in $\mathcal{V}_{h,0}$ closest to u in the energy norm.

A weighted best-approximation example

Let

$$V = \mathbb{R}^2, \qquad V_h = \operatorname{span}\{(1,1)\}, \qquad \boldsymbol{u} = (1,0),$$

and define

$$a(\boldsymbol{x}, \boldsymbol{v}) = 4x_1 v_1 + x_2 v_2, \qquad \|\boldsymbol{x}\|_a = \sqrt{4x_1^2 + x_2^2}.$$

Every vector in V_h has the form $\boldsymbol{v}_h = t(1,1)$. Therefore

$$\|\boldsymbol{u} - \boldsymbol{v}_h\|_a^2 = 4(1-t)^2 + t^2.$$

Differentiating this quadratic with respect to t gives

$$-8(1-t) + 2t = 0, \qquad t = \frac{4}{5}, \qquad \boldsymbol{u}_h = \left(\frac{4}{5}, \frac{4}{5}\right).$$

The Euclidean closest point is $\frac{1}{2}(1,1)$. The discrete space is the same, but the norm changes the best approximation.

Céa's lemma

Let V be a Hilbert space and $V_h \subset V$ a finite-dimensional subspace. Assume that

$$|a(w,v)| \leq M \|w\|_V \|v\|_V , \qquad a(v,v) \geq \alpha \|v\|_V^2 .$$

If u and u_h solve the continuous and discrete Galerkin problems, then

$$\boxed{\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V .}$$

The estimate compares the computed error with the smallest error attainable in V_h.

Proof of Céa's lemma: coercivity and orthogonality

Let $e = u - u_h$ and choose any $v_h \in V_h$. Coercivity gives

$$\alpha \|e\|_V^2 \leq a(e, e).$$

Because

$$e = (u - v_h) + (v_h - u_h)$$

and a is bilinear in its second argument,

$$a(e, e) = a(e, u - v_h) + a(e, v_h - u_h).$$

Because $v_h - u_h \in V_h$, Galerkin orthogonality gives

$$a(e, v_h - u_h) = 0.$$

Hence

$$\alpha \|e\|_V^2 \leq a(e, u - v_h).$$

Proof of Céa's lemma: continuity

Continuity of a gives

$$\alpha \|e\|_V^2 \leq M \|e\|_V \|u - v_h\|_V .$$

If $e \neq 0$, divide by $\alpha \|e\|_V$:

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \|u - v_h\|_V .$$

This holds for every $v_h \in V_h$. Taking the infimum gives Céa's estimate.

For diffusion in the H^1 norm,

$$\frac{M}{\alpha} = \frac{k_1}{k_0}(1 + C_P^2).$$

This factor does not depend on h or p.

Finite element interpolant

To use Céa's lemma, we need a specific function $v_h \in V_h$ whose error can be estimated. We choose the nodal interpolant.

Let V_h be a Lagrange finite element space with global nodes $\boldsymbol{x}_i$ and nodal basis functions ϕ_i. For a continuous function u, define

$$\boxed{\mathcal{I}_h u = \sum_{i=1}^{N_n} u(\boldsymbol{x}_i)\phi_i.}$$

The interpolant belongs to V_h and satisfies

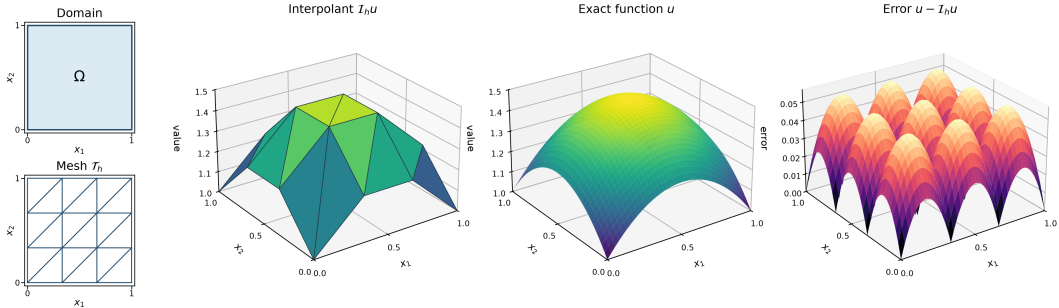
$$\mathcal{I}_h u(\boldsymbol{x}_i) = u(\boldsymbol{x}_i) \qquad i = 1, \ldots, N_n.$$

On an element $\mathcal{K}_e$,

$$(\mathcal{I}_h u)|_{\mathcal{K}_e} = \mathcal{I}_e(u|_{\mathcal{K}_e}).$$

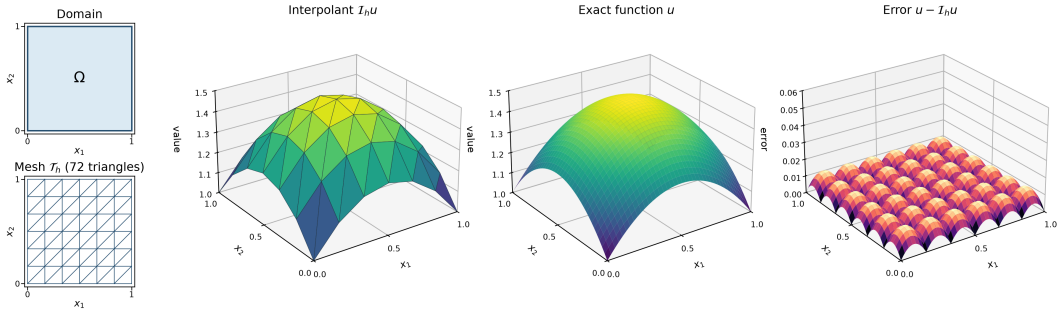
Thus the global interpolant can be constructed element by element from the nodal values of u.

Interpolation on a triangular mesh



The coarse mesh defines the P_1 interpolant. The finer plotting points only display the exact function and the error smoothly; they are not additional finite element nodes. For this quadratic and uniform mesh, the error pattern repeats on every triangle up to translation or reflection.

Interpolation after uniform refinement



Halving the grid spacing increases the mesh from 18 to 72 triangles. With the same vertical scales in both figures, the maximum error decreases from 1/18 to 1/72, a factor of four.

Properties and regularity of the interpolant

Three properties will be used in the error estimates.

1. **Locality:** $(\mathcal{I}_h u)|_{\mathcal{K}_e}$ depends only on the degrees of freedom of $\mathcal{K}_e$.
2. **Polynomial reproduction:** if $u|_{\mathcal{K}_e}$ belongs to the local finite element space, then $\mathcal{I}_e u = u$ on $\mathcal{K}_e$.
3. **Boundary values:** if $u = 0$ on Γ_D and the mesh resolves Γ_D, then $\mathcal{I}_h u \in \mathcal{V}_{h,0}$.

The nodal interpolant requires point values. In one dimension, H^1 functions are continuous. In two and three dimensions, a general H^1 function need not be continuous, so the estimates will assume additional regularity, commonly $u \in H^2(\Omega)$.

A four-element linear mesh

Consider $u(x) = 1 + x(1 - x)$ on $[0, 1]$. Use four uniform linear elements with five nodes:

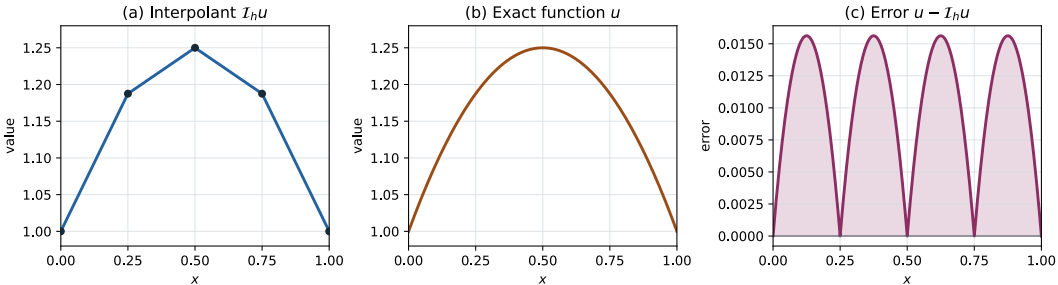
$$x_1 = 0, \qquad x_2 = \frac{1}{4}, \qquad x_3 = \frac{1}{2}, \qquad x_4 = \frac{3}{4}, \qquad x_5 = 1.$$

On $\mathcal{K}_e = [x_e, x_{e+1}]$,

$$\mathcal{I}_e u = u(x_e)\phi_1^e + u(x_{e+1})\phi_2^e.$$

The interpolant agrees with u at both element nodes and is linear between them. Adjacent local interpolants share the same value at their common node, so the global interpolant is continuous.

Exact function, interpolant, and error

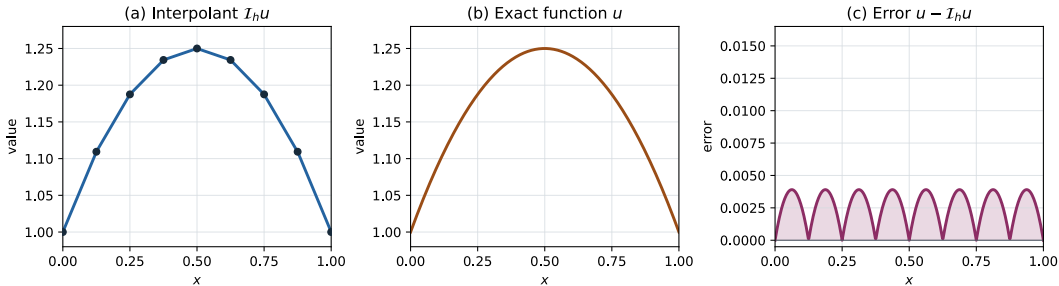


On every element,

$$u(x) - \mathcal{I}_e u(x) = (x - x_e)(x_{e+1} - x).$$

Since $h = 1/4$ on each element, the error reaches the same maximum $h^2/4 = 1/64$ at every element midpoint.

Interpolation on eight linear elements



Dividing every coarse element in two gives nine nodes $x_i = (i-1)/8$. The element length decreases from $1/4$ to $1/8$, and the maximum error decreases from $1/64$ to $1/256$.

Local interpolation estimates

Let $\mathcal{K}_e = (x_e, x_{e+1})$, $h_e = x_{e+1} - x_e$, and $u \in H^2(\mathcal{K}_e)$. Then

$$\boxed{\left\|(u - \mathcal{I}_e u)'\right\|_{L^2(\mathcal{K}_e)} \leq h_e \left\|u''\right\|_{L^2(\mathcal{K}_e)}}$$

and

$$\boxed{\left\|u - \mathcal{I}_e u\right\|_{L^2(\mathcal{K}_e)} \leq h_e^2 \left\|u''\right\|_{L^2(\mathcal{K}_e)}.}$$

The derivative error contains one power of h_e. The error in the function values contains two powers of h_e.

Derivative estimate on one element

Set

$$w = u - \mathcal{I}_e u.$$

Because $\mathcal{I}_e u$ is linear, $w'' = u''$. Also,

$$w(x_e) = w(x_{e+1}) = 0.$$

Rolle's theorem gives $x_* \in (x_e, x_{e+1})$ such that $w'(x_*) = 0$. Hence

$$w'(y) = \int_{x_*}^{y} w''(t)\, \mathrm{d}t.$$

Cauchy–Schwarz gives

$$|w'(y)|^2 \leq h_e \left\|w''\right\|^2_{L^2(\mathcal{K}_e)}.$$

Integrating over an interval of length h_e yields

$$\left\|w'\right\|_{L^2(\mathcal{K}_e)} \leq h_e \left\|u''\right\|_{L^2(\mathcal{K}_e)}.$$

L^2 estimate on one element

Use $w(x_e) = 0$ and write

$$w(y) = \int_{x_e}^{y} w'(t)\ \mathrm{d}t.$$

Cauchy–Schwarz and integration over $\mathcal{K}_e$ give

$$\|w\|_{L^2(\mathcal{K}_e)} \leq h_e \left\|w'\right\|_{L^2(\mathcal{K}_e)}.$$

Combining this relation with the derivative estimate gives

$$\|u - \mathcal{I}_e u\|_{L^2(\mathcal{K}_e)} \leq h_e^2 \left\|u''\right\|_{L^2(\mathcal{K}_e)}.$$

The two powers of h_e arise from two applications of the fundamental theorem of calculus and the Cauchy–Schwarz inequality.

From local estimates to global estimates

Because the elements have disjoint interiors,

$$|u - \mathcal{I}_h u|^2_{H^1(0,L)} = \sum_{e=1}^{N} \left\| (u - \mathcal{I}_e u)' \right\|^2_{L^2(\mathcal{K}_e)},$$

and

$$\|u - \mathcal{I}_h u\|^2_{L^2(0,L)} = \sum_{e=1}^{N} \|u - \mathcal{I}_e u\|^2_{L^2(\mathcal{K}_e)}.$$

Applying the local bounds gives

$$|u - \mathcal{I}_h u|^2_{H^1(0,L)} \leq \sum_{e=1}^{N} h_e^2 \left\| u'' \right\|^2_{L^2(\mathcal{K}_e)},$$

$$\|u - \mathcal{I}_h u\|^2_{L^2(0,L)} \leq \sum_{e=1}^{N} h_e^4 \left\| u'' \right\|^2_{L^2(\mathcal{K}_e)}.$$

What to retain from today

- Continuity and coercivity provide a stable continuous and discrete problem.
- Galerkin orthogonality states that $a(u - u_h, v_h) = 0$ for every $v_h \in \mathcal{V}_{h,0}$.
- Céa's lemma bounds the error by the best approximation available in that space.
- For symmetric problems, the finite element solution is the best approximation in the energy norm.

The next lecture constructs the nodal interpolant, derives interpolation estimates, and combines them with Céa's lemma to obtain finite element error estimates.